FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, a farm manager or a system administrator, I want one central Salesforce platform where farm, crop, farmer and buyer details are stored safely and in an organised way. Entering and updating records should be easy through a simple interface, and validation rules should stop wrong entries before they are saved. I also want real-time reports and interactive dashboards so I can track farm productivity, watch sales and take better decisions. Finally, role-based access and secure storage should keep sensitive information protected while still letting the team work together. Overall, the system should reduce paperwork and mistakes and make farm management faster and more reliable.

1. INTRODUCTION

1.1 Project Overview

The Farm Management System (FMS) is a Salesforce application built to move day-to-day farming records from paper registers into one digital platform. It keeps track of farms, crops, farmers and buyers, and turns that data into reports and dashboards that farmers and managers can read at a glance. With this, they can follow agricultural activities closely and decide with confidence.

1.2 Purpose

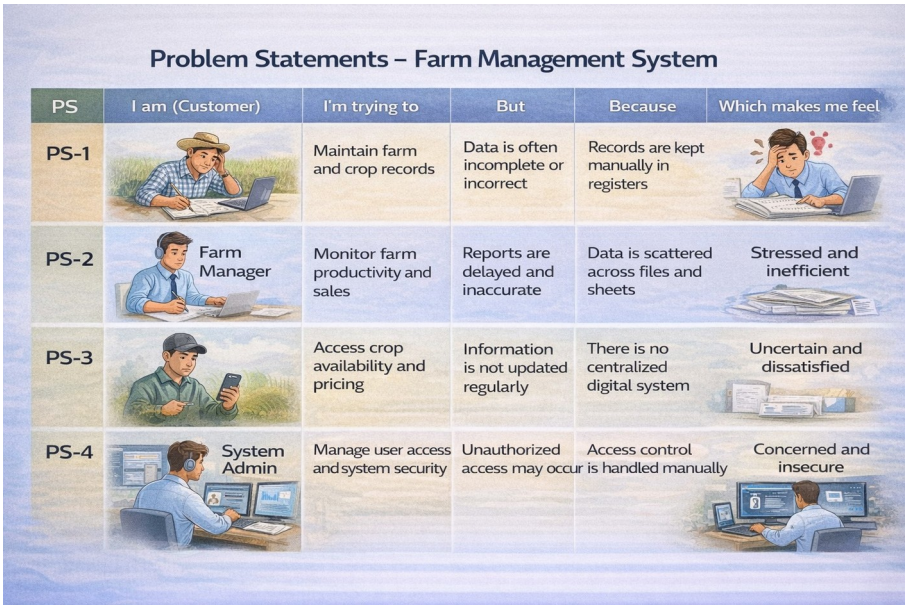
The main aim of this project is to:

- Move agricultural records from paper to a digital system
- Keep data accurate and easy to access
- Reduce manual record handling through automation
- Allow real-time monitoring of farms and crops
- Support decisions with data instead of guesswork
- Improve productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure



2.2 Empathy Map Canvas

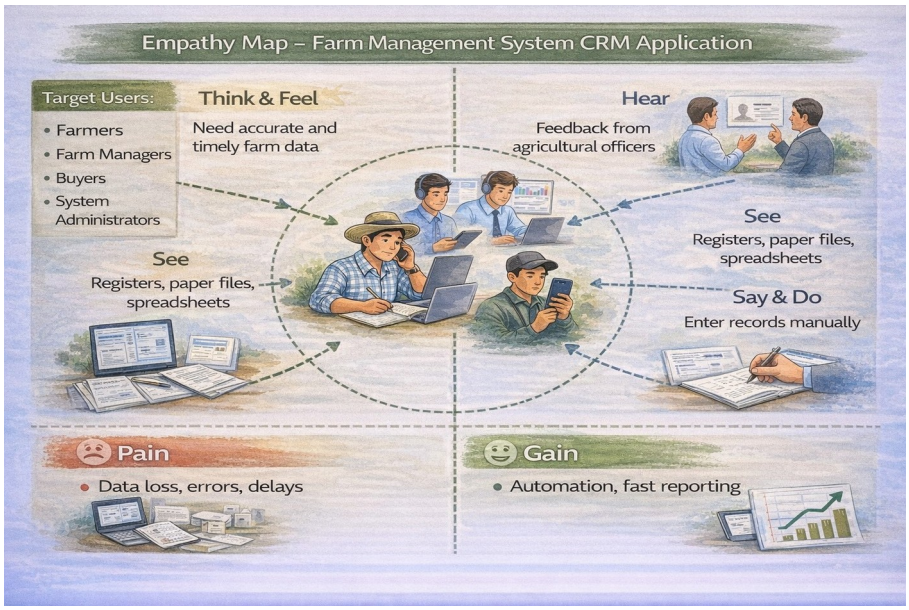
Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- Think & Feel: Need accurate and timely farm data
- Hear: Feedback from agricultural officers
- See: Registers, paper files, spreadsheets

- Say & Do: Enter records manually
- Pain: Data loss, errors, delays
- Gain: Automation, fast reporting



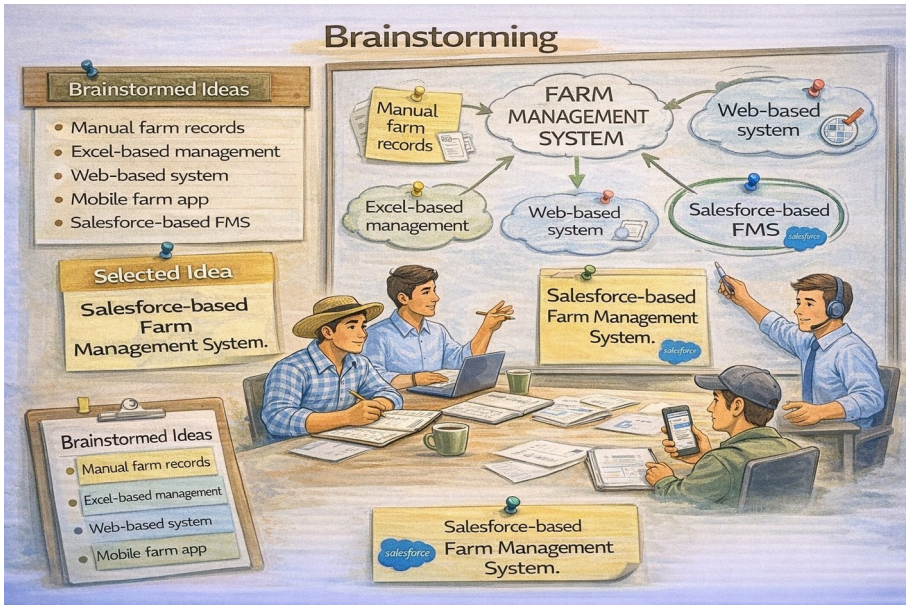
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

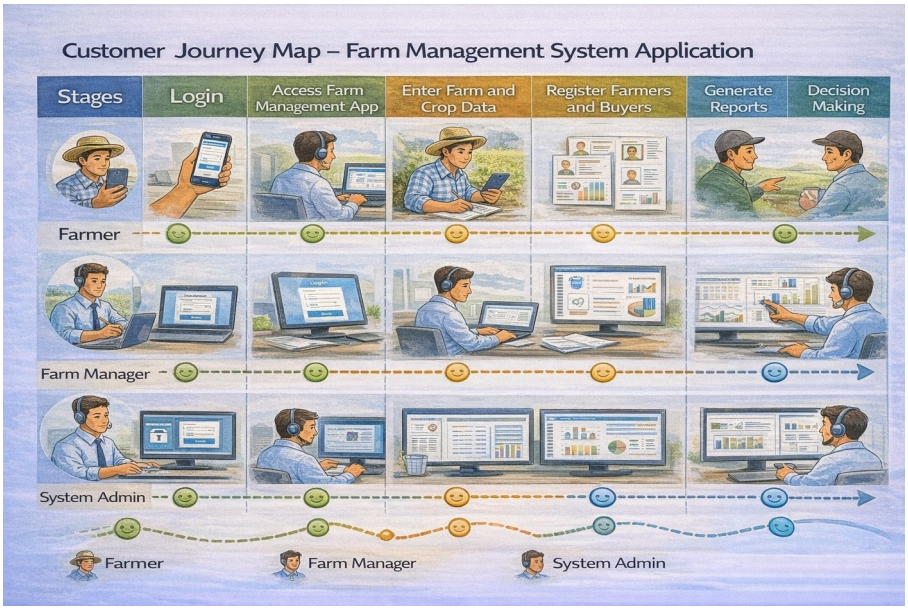


3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

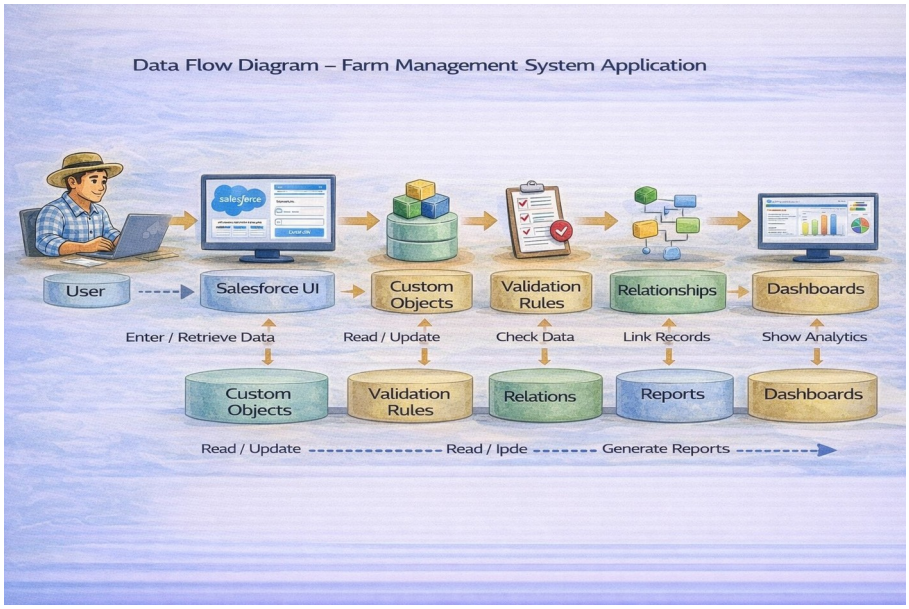
FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow: User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



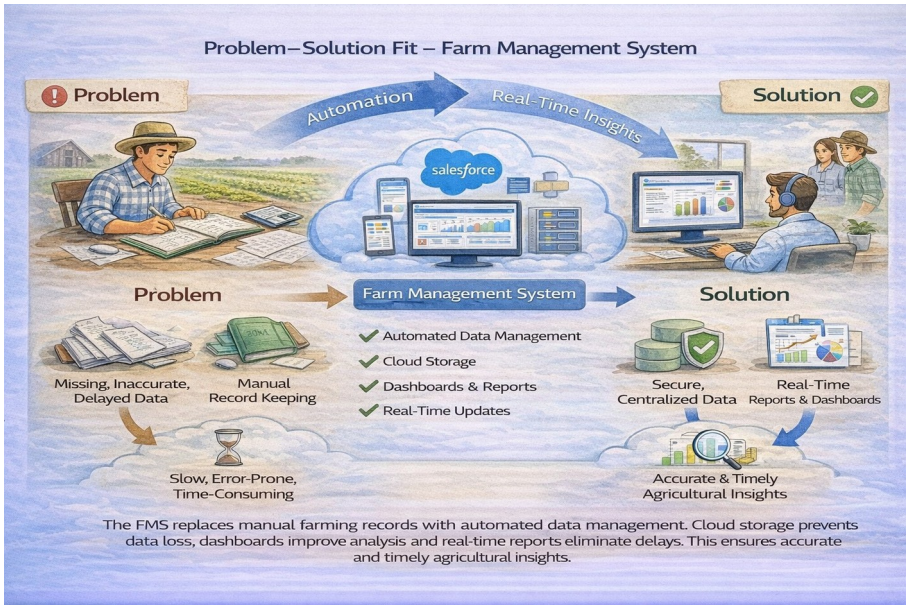
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

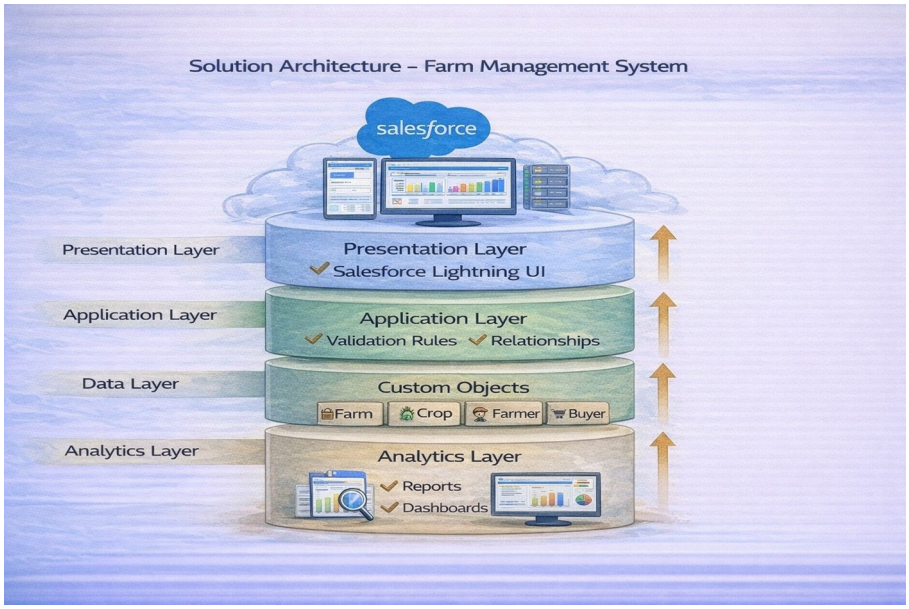
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.
Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

- Presentation Layer – Salesforce Lightning UI
- Application Layer – Validation Rules, Relationships
- Data Layer – Custom Objects

- Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	20	6 Days	31 Aug 2026	05 Sep 2026	20	05 Sep 2026
Sprint-2	20	6 Days	07 Sep 2026	12 Sep 2026	20	12 Sep 2026
Sprint-3	20	6 Days	14 Sep 2026	19 Sep 2026	20	19 Sep 2026
Sprint-4	20	6 Days	21 Sep 2026	26 Sep 2026	20	26 Sep 2026

6. PROJECT DEVELOPMENT PHASE

Project Description:

The Farm Management System (FMS) in Salesforce is a complete solution to digitize and streamline agricultural operations. It helps farmers, farm managers and other stakeholders manage crop, farm, farmer and buyer data, and gives them useful insights through reports and dashboards.

Milestone 1 – Salesforce

What is Salesforce?

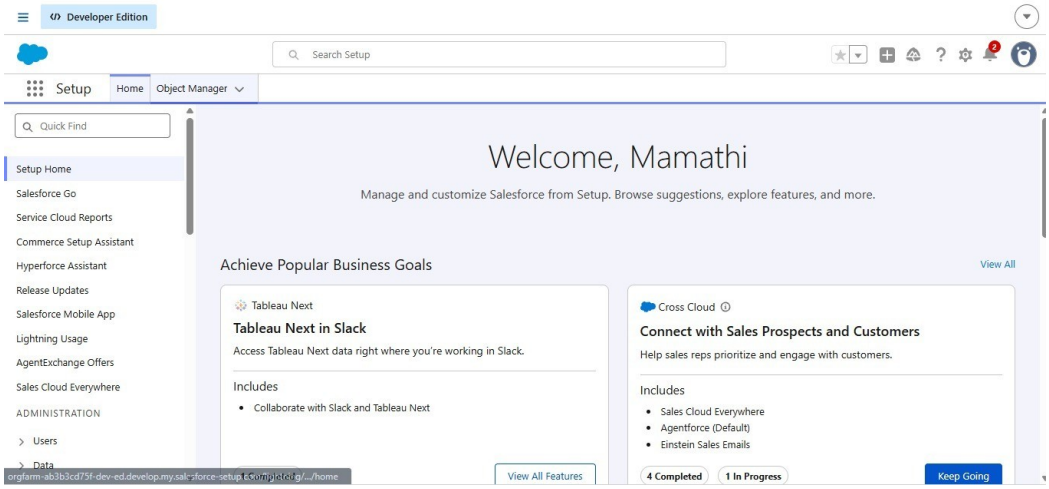
Salesforce is a customer success platform designed to help you sell, service, market, analyze and connect with your customers. Using its standard products and features, you can manage

relationships with prospects and customers, collaborate with employees and partners, and store your data securely in the cloud.

Activity 1: Creating a Salesforce Developer Org

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Go to developer.salesforce.com/signup.
2. Enter the details: First name, Last name, Email, Role, Company, Country/Region, Postal code, and a unique Username.
3. Click Sign me up. After a few minutes you will receive a mail; use the verify account link to create your new password.
4. Click Save.
5. Go to login.salesforce.com.
6. Log in with your username and password. The Setup home page will appear.



Milestone 2 – Object

What is an object?

Salesforce objects are database tables that let you store data specific to an organization. An object consists of fields (columns) and records (rows).

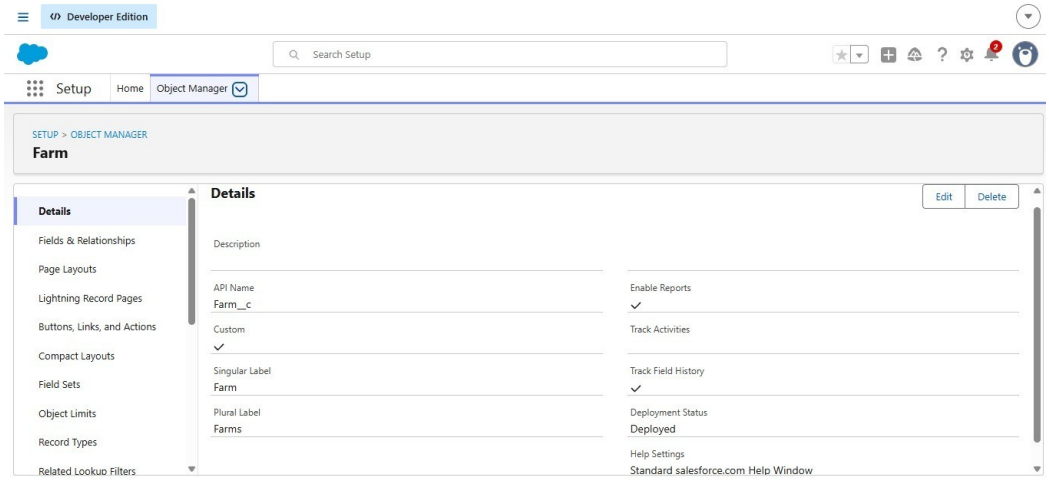
In this application we use 4 custom objects: Farm, Crop, Farmer and Buyer.

Activity 1: Create a custom object for Farm

1. From Setup, click Object Manager.
2. Click Create and select Custom Object.
3. Fill in the label as "Farm".
4. Fill in the plural label as "Farms".
5. Record name: Farm Name.
6. Select the data type as "Text".
7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is, and click Save.



Activity 2: Creation of Crop object

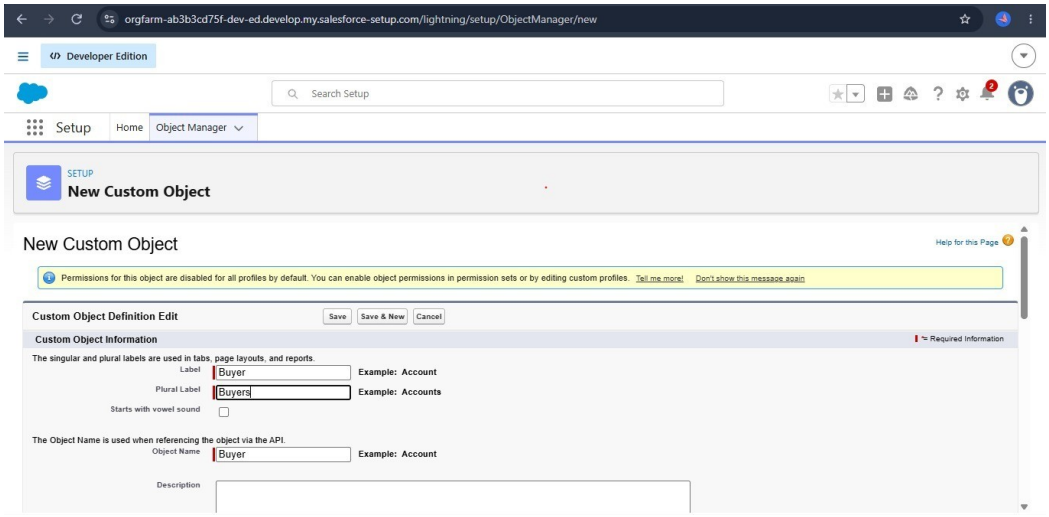
1. Click the gear icon and select Setup.
2. Click the Object Manager tab beside the Home tab.
3. Click the Create dropdown at the extreme right and select Custom Object.
4. Label: Crop. Plural Label: Crops. Record Name: Crop Name. Data type: Text.
5. Check Allow Reports and Allow Search.
6. In Object Creation Options, select Add Notes and Attachments related list to default page layout.
7. Click Save.

Activity 3: Creation of Farmer object

1. Click the gear icon and select Setup, then open Object Manager.
2. Click Create and select Custom Object.
3. Label: Farmer. Plural Label: Farmers. Record Name: Farmer Name. Data type: Text.
4. Check Allow Reports and Allow Search.
5. Select Add Notes and Attachments related list to default page layout, then click Save.

Activity 4: Creation of Buyer object

1. Click the gear icon and select Setup, then open Object Manager.
2. Click Create and select Custom Object.
3. Label: Buyer. Plural Label: Buyers. Record Name: Buyer Name. Data type: Text.
4. Check Allow Reports and Allow Search.
5. Select Add Notes and Attachments related list to default page layout, then click Save.



Milestone 3 – Tab

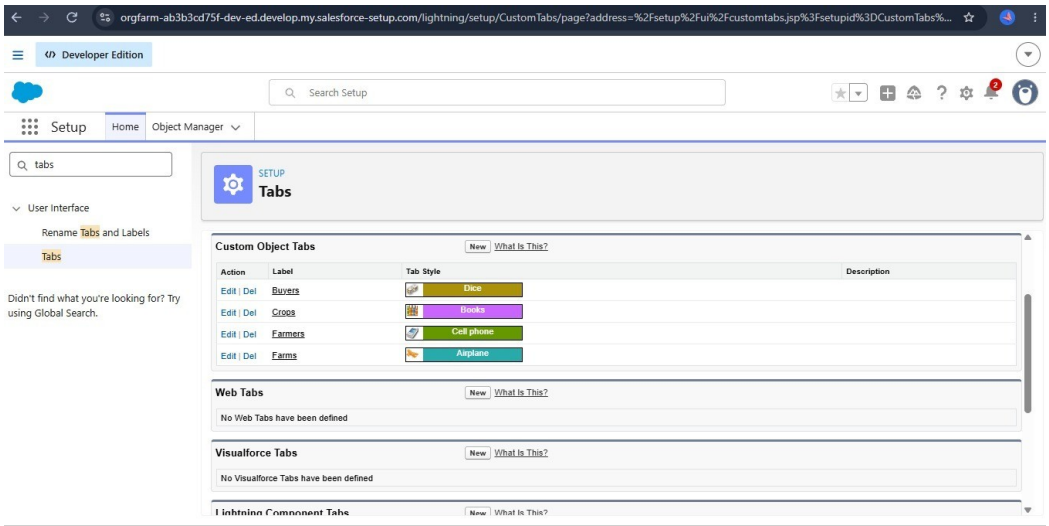
Tabs in Salesforce help users view information at a glance. They display the data of objects and other web content in the application.

Activity 1: Creation of Farm tab

1. Click the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activities 2, 3 and 4: Crop, Farmer and Buyer tabs

Repeat the same steps and select the Crop, Farmer and Buyer objects one by one. After this, all four custom object tabs appear in the list.



Milestone 4 – Lightning App

What is an App?

An app in Salesforce is a group of tabs that work together as a unit. It has a name, a logo and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create the Farm Management System app

- From Setup, enter App Manager in Quick Find and select App Manager.
- Click New Lightning App. Enter Farm Management System as the App Name, then click Next.
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer, Reports and Dashboards and move them to Selected Items. Click Next.
- From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
- To verify, click the App Launcher, type Farm Management System and select the app.

Note: the App Launcher displays the available apps, the App Name shows the currently selected app, and the Navigation menu displays the tabs available inside the app.

Milestone 5 – Fields and Relationships

What are fields?

Fields in Salesforce are like the columns of a relational database. They store the data values required for a particular object in a record.

Activity 1: Creation of fields for the Farm object

1. Click the gear icon and select Setup. This opens Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation and click New.
5. Choose the data type Picklist and click Next.
6. Enter the Field Label (Farm Type) and the values Crop, Livestock, Mixed.
7. Click Next, Next and Save.

Now create the other fields in the same way, choosing the data types carefully:

- Picklist, Field Label Soil Quality, values: Organic, Cover Crops, Composting
- Picklist, Field Label Irrigation Type, values: Sprinkler irrigation, Drip irrigation, Localized irrigation
- Text, Field Label Location
- Text, Field Label Farm Size (Acres)
- Master-Detail relationship with the Farmer object

Activity 2: Creation of fields for the Crop object

- Text, Field Label Crop Name

- Date, Field Label Planting Date
- Date, Field Label Harvest Date
- Lookup relationship from Crop to the Farm object
- Currency, Field Label Actual Yield
- Number, Field Label Total Acres Harvested, maximum length 20
- Formula field: select the data type Formula and click Next. Enter the Field Label Expected Yield per Acre and choose Currency as the return type. In the formula, divide Actual Yield by Total Acres Harvested (Actual_Yield__c / Total_Acres_Harvested__c), then click Next and Save.

Activity 3: Creation of fields for the Farmer object

- Phone, Field Label Contact Number
- Picklist, Field Label State, values: Telangana, Andhra Pradesh, Tamilnadu
- Picklist, Field Label District, with the district values
- Picklist, Field Label Village, with the village values

Creation of field dependency

1. Open Object Manager, choose the Farmer object and select Fields & Relationships. Click the Field Dependencies button on the right side.

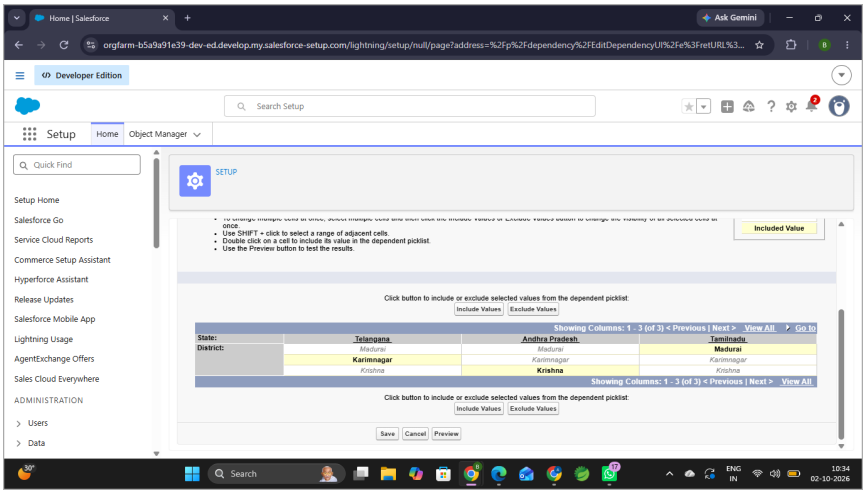
2. Click New. Select State as the Controlling Field and District as the Dependent Field, then click Continue.

3. Select the appropriate values in each column by double-clicking them. For example, Telangana – Yadadri Bhongir.

4. Click Include Values. Do the same for Andhra Pradesh and Tamilnadu with their districts.

5. Click Preview and test the dependency by choosing different states and checking the districts shown.

6. Click Close to close the preview window, then click Save.



Activity 4: Creation of fields for the Buyer object

- Phone, Field Label Contact
- Email, Field Label Email

- Text Area, Field Label Address
- Picklist, Field Label Buyer Type, values: Wholesale, Retail

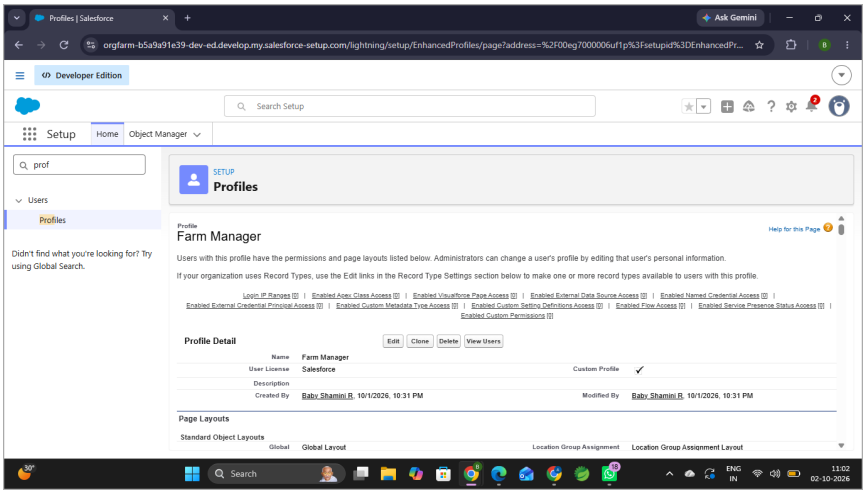
Milestone 6 – Profile

What is a profile?

A profile is a collection of settings and permissions that define what a user can do in Salesforce. It controls object permissions, field permissions, user permissions, tab settings, app settings, Apex class access, Visualforce page access, page layouts, record types, login hours and login IP ranges. A profile can be assigned to many users, but a user can have only one profile at a time.

Activity 1: Create a custom profile

1. From Setup, enter Profiles in the Quick Find box.
2. Select Profiles and open Standard User.
3. Click Clone.
4. For Profile Name, enter Sales Department.
5. Click Save.



Activity 2: Create a custom profile 2

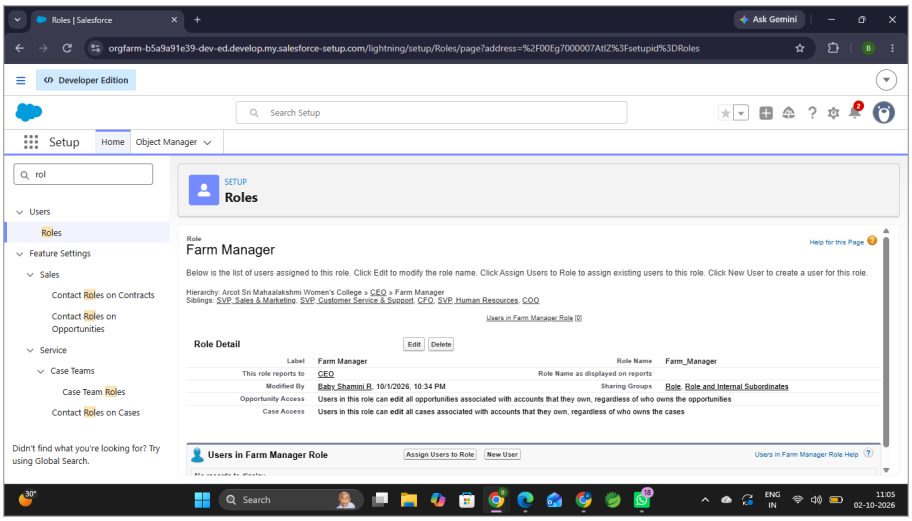
Create a profile named "Farm Manager" by cloning Standard User in the same way.

Milestone 7 – Role

In Salesforce, roles decide which users can see certain data and functions. They also define the reporting hierarchy of an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of role

1. From the Quick Find box, search for Roles and click the Roles option.
2. Select the Set Up Roles option.
3. Below the CEO, click Add Role, enter the label "Farm Manager" (the role name is filled in automatically) and click Save.



Milestone 8 – User

What is a user?

A user is anyone who logs into Salesforce. Users are employees such as sales reps, managers and IT specialists who need access to the company's records. Every user has a user account, and its settings decide which features and records the user can access.

Note: in a Developer Org a Salesforce license can be used by only 2 users at a time. If you cannot find the Salesforce license, deactivate a user who has it, or change the license type from Salesforce to another one.

Activity 1: Create a user

1. From Setup, enter Users in the Quick Find box and select Users.
2. Click New User.
3. Enter the first name as Farm and the last name as Manager.
4. Enter the email address and a unique username in email format (by default it is the same as the email).
5. Select the role Farm Manager.
6. Select the User License as Standard Platform User.
7. Select the profile Farm Manager.
8. Click Save.

Activity 2

Create another user with the name "Sales Department" and assign the Sales Manager profile, following the same steps.

Milestone 9 – Sharing Rules

What are sharing rules?

Sharing rules let users share records based on conditions. They are used for objects whose organization-wide default (OWD) is Private or Public Read-Only, because sharing rules can only extend access, never restrict it. There are two types: owner-based and criteria-based sharing rules.

Activity 1: Create a sharing rule

1. Go to Sharing Settings by searching in Quick Find.
2. Scroll down to the Crop object and click New to create a sharing rule.
3. Add a label for the sharing rule.
4. Select the rule type as criteria-based.
5. Select the criteria field from the Crop object, set the operator to equals and give the value.
6. Under "Select the users to share with", choose Roles and select Farm Manager.
7. Under the access level, choose Read/Write.
8. Save the rule.

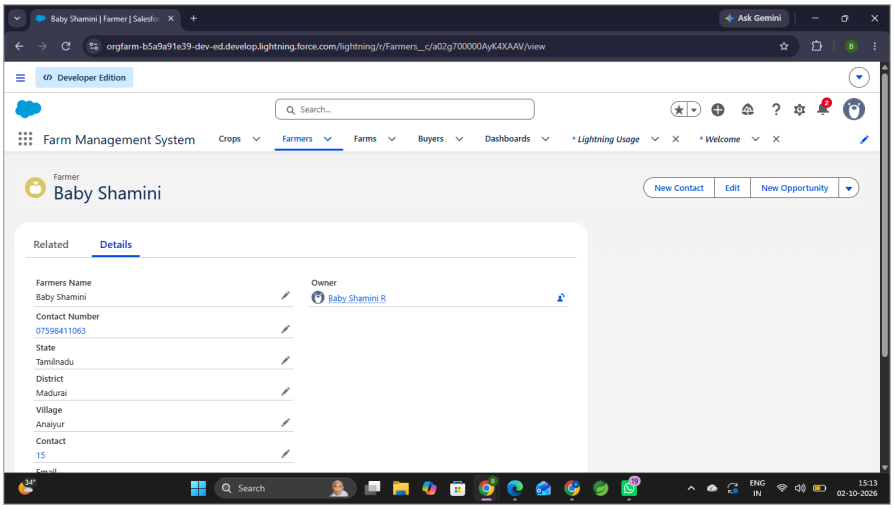
Activity 2

Create a sharing rule to share the Buyer records with the Sales Department with Read/Write access, following the steps above.

Milestone 10 – User Adoption

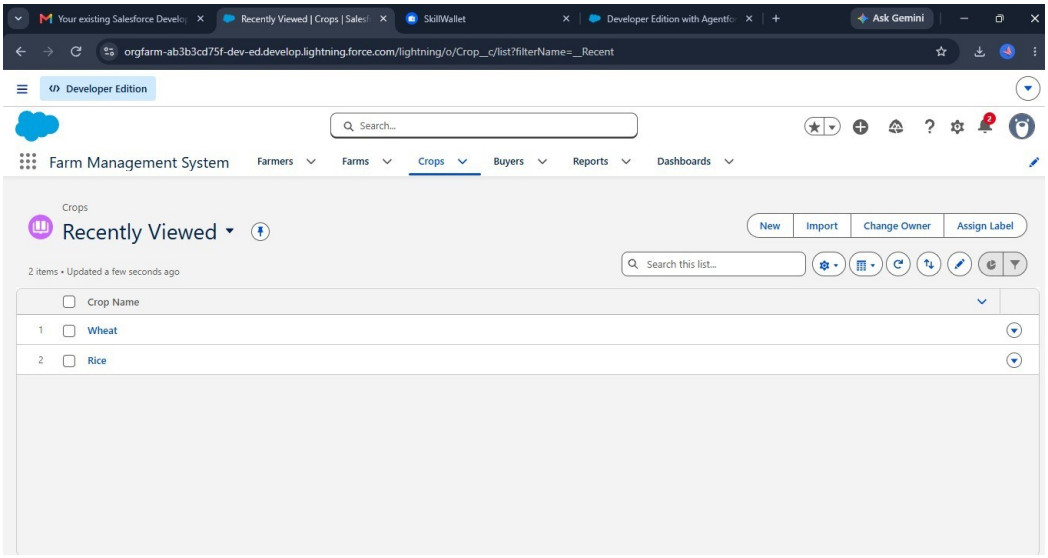
Activity 1: Create a record (Farmer)

- Click the App Launcher on the left side of the screen.
- Search for Farm Management System and click it.
- Click the Farmers tab.
- Click New, fill in the details and Save.



Activity 2: Create records (Crop)

In the same way, open the Crops tab, click New and save the crop details.



Milestone 11 – Reports

What are reports?

A report is a list of records that meet the criteria you define. It is shown in rows and columns and can be filtered, grouped or displayed as a chart. Every report is stored in a folder. Folders can be public, hidden or shared, and can be read-only or read/write.

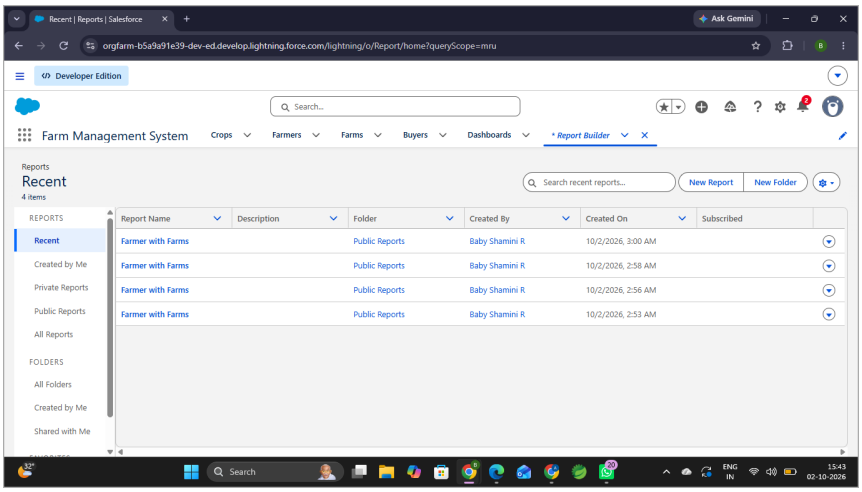
Activity 1: Create a report

Create a report that shows farmer names with their farms, irrigation type and farm size.

1) Click the App Launcher and search for Reports.
2) Click New Report and select the report type "Farmers with Farms".
3) In the Details section, click Start Report.
4) In the Outline, click Columns and select the fields you want to display.
5) Click Group and select the fields you want to group by.
6) Save the report with the label "Farmers with Farms" in a public folder.

Activity 2

Create reports using the Crops and Buyers objects, following the steps above.

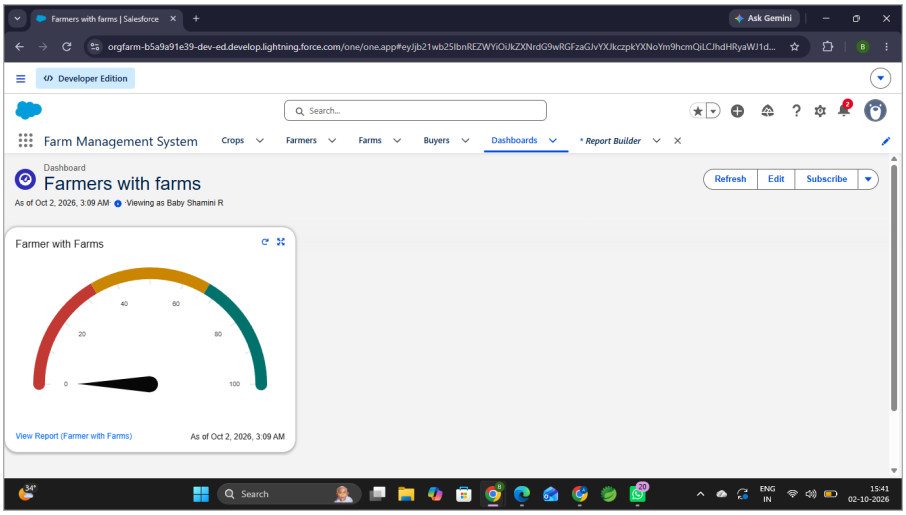


Milestone 12 – Dashboard

Dashboards give more insight than reports because they combine data from many reports and show a summarized result. Looking at many reports at once makes it quicker to decide on action plans. Dashboards can contain charts, graphs and tabular data.

Activity 1: Create a dashboard

1. Click the Dashboards tab.
2. Click New Dashboard.
3. Name the dashboard "Farmers with farms" and click Create.
4. Click + Component.
5. Select the Farmers with Farms report and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



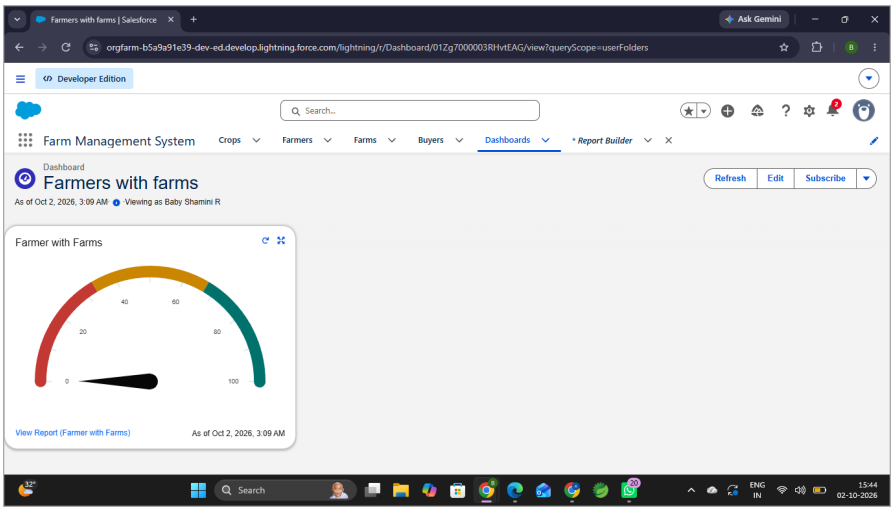
Milestone 13 – View Report and Dashboard

Activity 1: Report

- Click the App Launcher on the left side of the screen.
- Search for Farm Management System and click it.
- Click the Reports tab.
- Click Farmers with Farms and view the records.

Activity 2: Dashboard

- Click the App Launcher on the left side of the screen.
- Search for Farm Management System and click it.
- Click the Dashboards tab.
- Click Farmers with farms and view the records.



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from the Salesforce interface to verify that the system works as expected. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing confirmed smooth response and stable behaviour when several operations were done together.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. CONCLUSION

In this project we built a Farm Management System on Salesforce that replaces manual registers with one secure digital record. Using custom objects, relationships, profiles, roles,

sharing rules, reports and dashboards, the application makes data entry cleaner, reduces mistakes and gives managers a clear view of farms, crops and buyers. The project also showed us how a real-world problem can be solved on the Salesforce platform mostly through configuration, without heavy coding.

10. FUTURE SCOPE

The system can be extended further in the following ways:

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

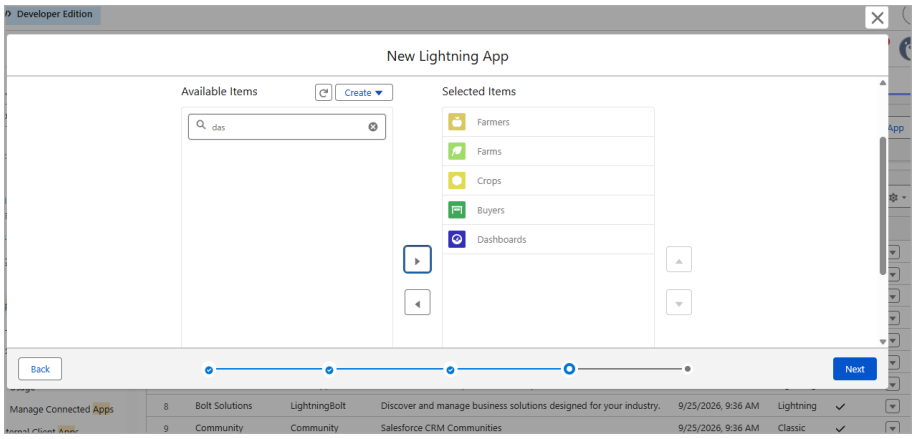
11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

ADDITIONAL SCREENSHOTS

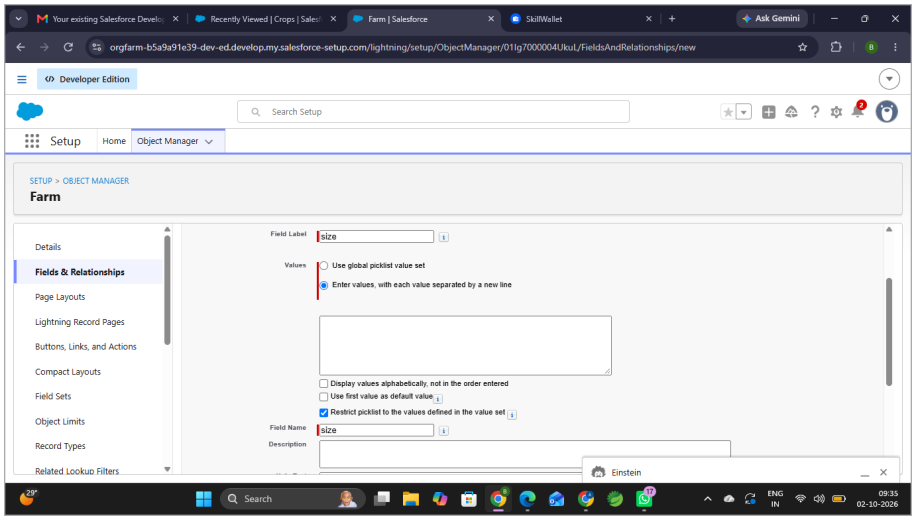
Milestone 4 – Lightning App

Farmers, Farms, Crops, Buyers and Dashboards added as selected items for the Farm Management System app

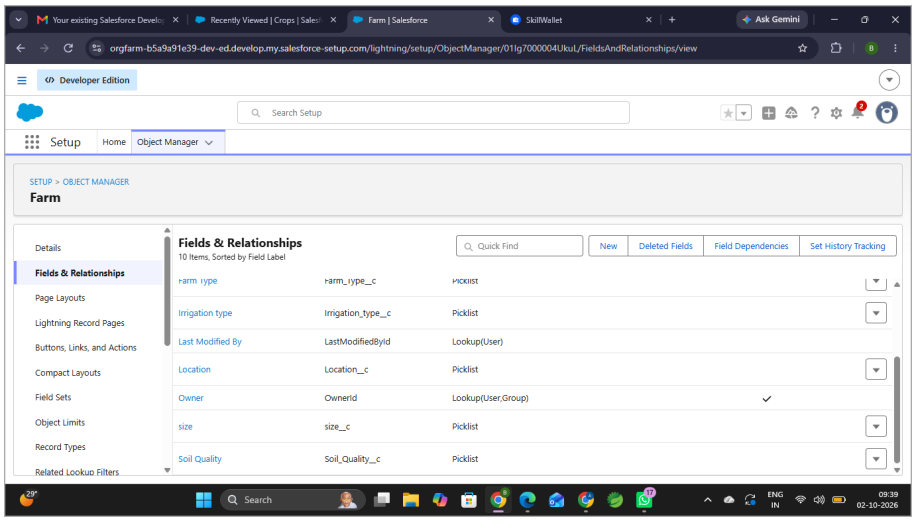


Milestone 5 – Fields and Relationships (Farm)

Farm object: creating the size picklist field

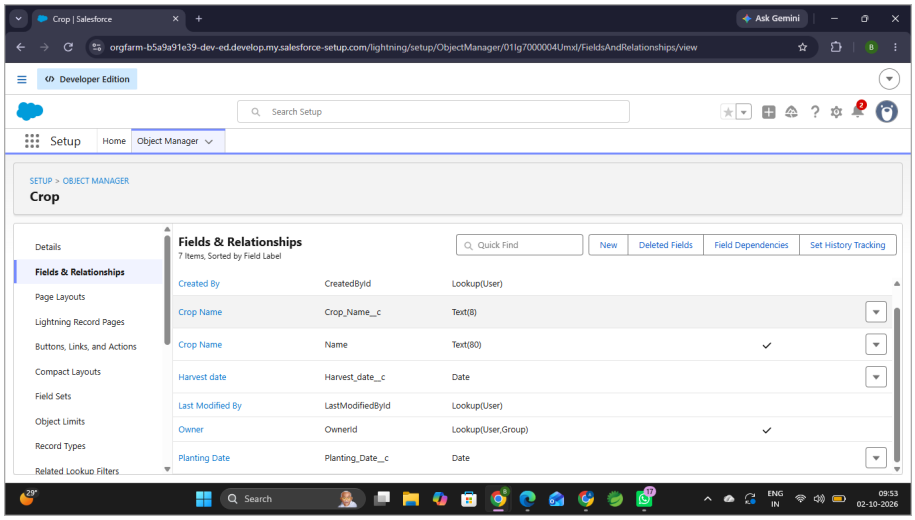


Farm object: Fields & Relationships list

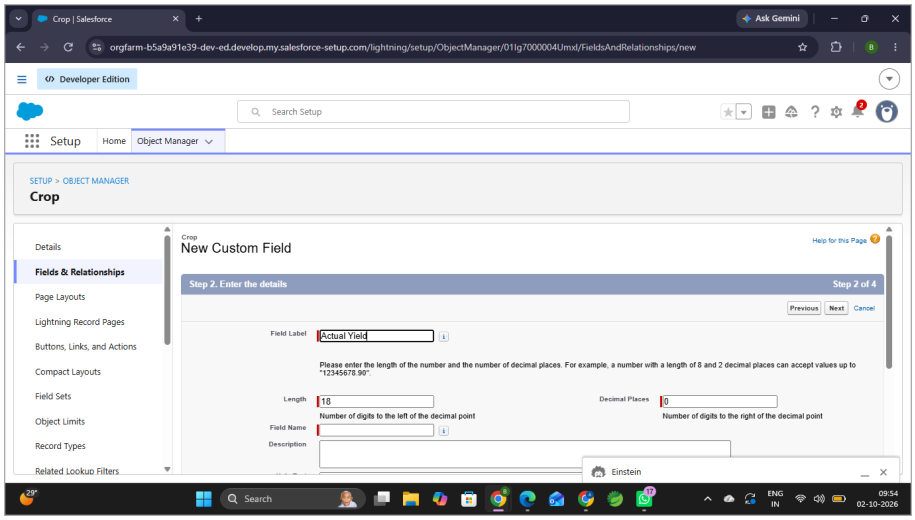


Milestone 5 – Fields and Relationships (Crop)

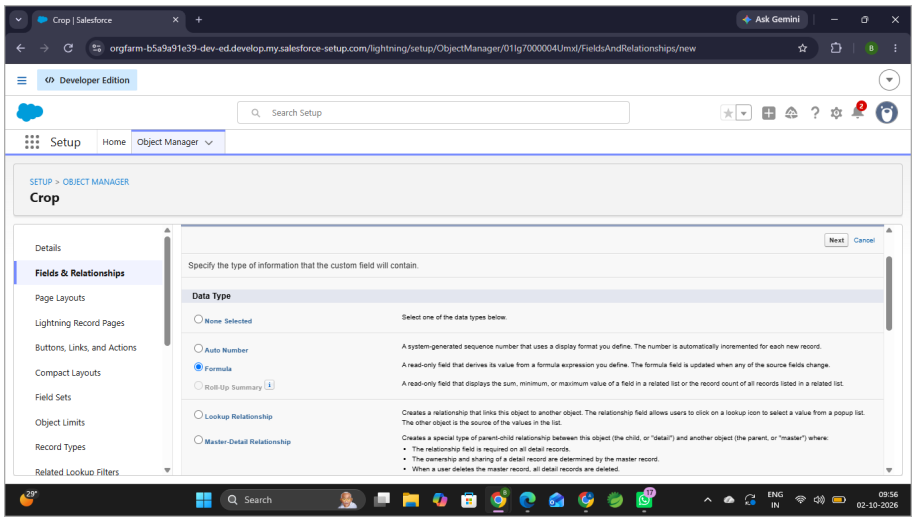
Crop object: Fields & Relationships list



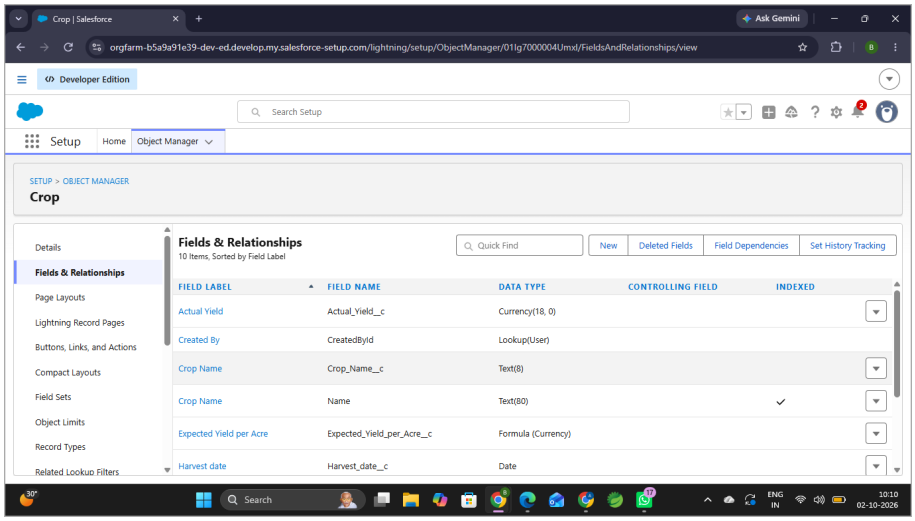
Crop object: creating the Actual Yield field



Crop object: selecting the Formula data type

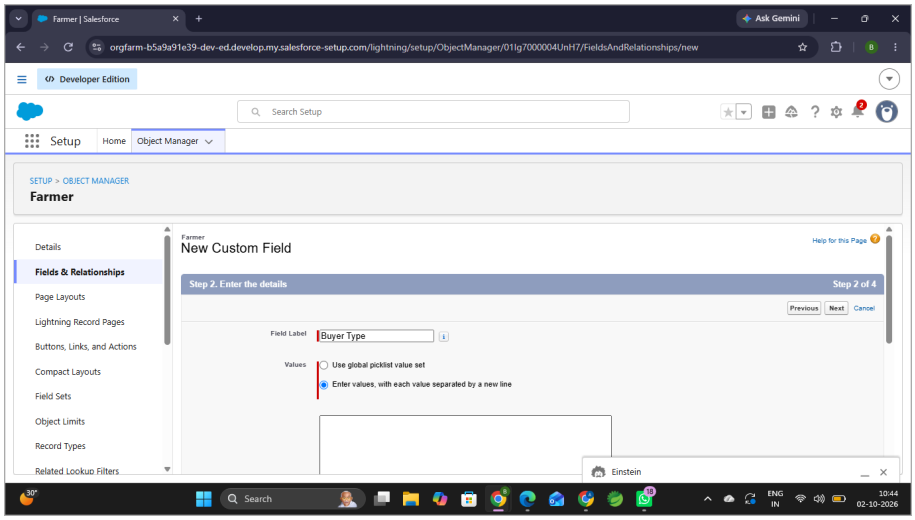


Crop object: fields list with the Expected Yield per Acre formula field



Milestone 5 – Fields and Relationships (Buyer Type)

Creating the Buyer Type picklist field (opened from the Farmer object)



Milestone 8 – User

Sales Manager user details

